



Instructor: Date:

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Let's Start!

Is a URL the same as a web address?



Learning Outcome:

Describe common simple incidents at work in a simple spoken or written situation with a supportive interlocutor. Assess the incident of data in the macrosystem processes, based on the data dictionary to consolidate intervening data, based in established parameters.

Can Do Statements:

- I can discuss what URL encoding is.
- I can list some types of URL characters accepted by web browsers and servers.
- I can talk about characteristics for different types of URL characters.

Ask your partner:

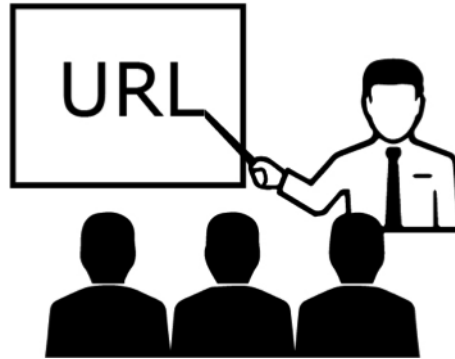
- How can websites get and answer users' requests?
- Do you know how to define the location of a computer network? List the steps.





Let's Explore!

An expert is lecturing about URL Encoding



Instructions: Read the text below.

Is there a specific way to request web pages? Absolutely, we can do that through URL encoding. It is a process in which is possible to transform characters into a specific format in order to transmit the data over the internet. A URL or a Uniform Resource Locator is the element that allows specifying the location of a computer network and the way to retrieve requests in a browser. URL's work mainly with pages that use the HTTP protocol but we can also use it for file transfer (ftp), email (mailto) or database access (JDBC).

Are there any specifications to set a URL? Well, most of URL's have to be displayed in an address bar above the webpage. They include basically three parts: a protocol, a hostname and a file name. For instance, if we have the site `https://en.wikiloop.org/Uniform_Resource_example.html` in that case the protocol will be https, the hostname "en.wikiloop.org", and the file name "Uniform_Resource_example.html"

Are there different types of characters in URL encoding? That's right. Servers and browsers can read and display different types of characters to ensure a universally accepted representation of data. There are mainly four types of characters. First, we have ASCII control characters, these are non-printing characters that we use for output control. They always include a code point or number within a character set. For instance, the number 127 is represented by the letters DEL. Then, we can also find Non-ASCII control characters, they make part of the ISO- Latin character set and they cover the set of characters beyond the ASCII character set of 128 characters.

In the third place, there are reserved characters. They make reference to special symbols such as comma, colon, semi - colon, forward slash, among others. As these types of characters may have different meanings in a URL, it is necessary to encode them.

Finally, there are some characters called unsafe because they can be misread within URL's; for this reason, they also require encoding. The pound character or the percentage characters are examples of this type of symbols.



**Let's Practice!**

I have a couple of questions about URL encoding.

Instructions: Match the sentences from Column A with Column B.

COLUMN A

1. Is there a specific way to request web pages?
2. Is there a definition of URL encoding in that text?
3. Are there any specific functions of a URL?

COLUMN B

- a. A URL or a Uniform Resource Locator allows specifying the location of a computer network and the way to retrieve requests in a browser.
- b. Absolutely, we can do that through URL encoding.
- c. It is a process in which is possible to transform characters into a specific format in order to transmit the data over the internet.

**Let's Practice!**

Are there any specifications to set a URL?

Instructions: Select the correct option to correctly complete each sentence.

There (am / is / are) different types of characters in URL encoding so that servers and browsers can (read / reads / reading) and display different types of symbols to ensure a universally accepted representation of data. To encode data, a programmer (use / uses / using) 4 types of characters: ASCII control characters, Non-ASCII control characters, Reserved characters, and Unsafe characters.





Extension Activity

I will present the character types for URI.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Make a list of ASCII characters.
- Search if they have special functions within the URL syntax.
- Discuss with a partner the functions of special characters.
- Make a brochure in which you define and present the character types present in URL encoding.

Self-Assessment

Yes

No

- I can identify and list ASCII characters.
- I can define the functions for special characters in URL encoding.
- I can explain the types of characters used in URL encoding.
- I can summarize the information about character types in URL encoding.

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Target Words

- protocol** (noun): a group of rules commanding the formats of exchangeable messages between computers
e.g. URL's work mainly with pages that use the HTTP protocol but we can also use it for file transfer (ftp), email (emailto) or database access (JDBC).
- encoding** (verb): to convert information into codes
e.g. Are there different types of characters in URL encoding?
- misread** (verb): to understand something in a wrong way
e.g. There are some characters called unsafe because they can be misread within URL's; for this reason, they also require encoding.
- retrieve** (verb): to recover something
e.g. A URL or a Uniform Resource Locator is the element that allows specifying the location of a computer network and the way to retrieve requests in a browser.
- address bar** (noun) the location bar of a web browser
e.g. Most of URL's have to be displayed in an address bar above the webpage.



Learning Points

There is and There are in Questions

You use **there is** and **there are** to indicate that an object or a person is located in a specific place or to say that something exists.

We use **there is** for singular uncountable nouns:

There is only a way to solve this coding problem.

There isn't any way to solve this coding problem.

Is there any way to solve this coding problem?

We use **there are** for plural countable nouns:

There are specific mechanisms to program a webpage.

There aren't specific mechanisms to program a webpage

Are there specific mechanisms to program a webpage?

To ask a question, we need to change the order of words in the sentence, this is the structure:

Is/Are + there + noun + complement?

Examples:

Is there an encoding language in URL?

Are there characters that need to be encoded in URL?

Examples

- *Is there a reference to a web resource you know?*
- *Is there a widely understood encoding for characters?*
- *Is there a way to encode characters?"*





Learning Points

Syntax for Uniform Resource Locator, URL

There are many factors involved in the syntax of a URL.
The syntax of a URL has the following items:

Syntax of a Generic URL	
Scheme	A set of characters that start with a letter and are followed by a combination of digits, letters, plus (+), periods (.), or hyphens (-). After a scheme, there is a colon as in "http:"
Two Slashes (//)	Slashes are required just by some schemes; if the authority component is absent, the two slashes are not necessary.
Authority Part	This element requires an authentication part and a host. In the first part, there is a username and a password separated by a colon and a "at" symbol (@). And in the other, you can find the IP address in brackets ([]).
Path	It contains data organized in a hierarchical way separated by slashes (/). The path starts with a single slash.
Query (Optional)	It contains non-hierarchical data and the information is preceded by a question mark (?).
Fragment (Optional)	It contains a fragment identifier that can link the information to another resource. In a fragment, the information is preceded by a hash (#).

This is an example of the syntax of a URL:

http://[user:password@]host[:port]][/]path[?query][#fragment]

